ML Based Automated Pharmacokinetics Modeling in Personal Medicine

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Abstract

Pharmokinetics research requires numerous amounts of data collection based on a patient's drug intake and its concentration level over a period of time. The explanatory model of this data is known as a PK model. However, current data analysis methods are mostly performed manually in the clinical study which is the post drug usage study. The real-time PK study in clinical practice is far from a viable option, since the labor and time cost is high, As such, we introduce the use of machine learning to automate the PK analysis in which the determination of model type, parameter fitting and detection of abnormalities are performed. Using Gaussian process (GP) regression for interpolating and modeling with K-nearest neighbours (KNN), detecting the abnormalities Density based clustering method(DBSCAN) we are able to predict accurate results for the type of drug dosage being used and its concentration levels.

1 Introduction

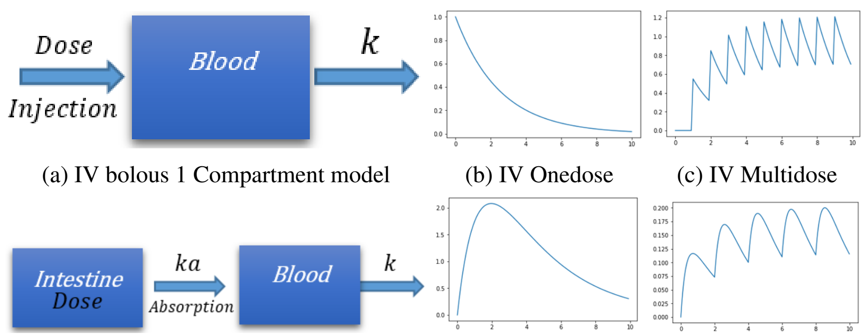
The pharmacokinetics (PK) study is a crucial part in the pre-clinical and clinical research. Before the drug is brought into the market, there are critical variables such as the half-life ($t^{\frac{1}{2}}$), elimination coefficient (k), absorption coefficient (k_a), minimum effect dose and minimum toxicity dose which must be determined. Conventionally, these information will be revealed based on the group of the patients enrolled in the clinical study of a drug. Once the kinetics coefficients are determined, the dosage form and the dosing intervals will be determined and included the drug's label. However, because of the limited sample size and the standardized drug administration method of the enrolled patients in the clinical study, the dosing regimen is not optimized for each individual. This introduces the problem of overdose or under-dose.

PK modeling is used to illustrate the time versus drug concentration correlation. The model assumes the organ and tissue are treated as separate compartments connected by influx or outflow of the drug. The rate is controlled by rate constant and the corresponding ordinary differential equations shown in Figure 1. The different types of PK models in Figure 1 varies based on different methods of administration. IV stand for intravenous injection shown in Eq.(1), in which the whole dosage of the drug will be injected into the blood circulation. PO stands for oral administration shown in Eq.(2). In this case, the drug will be absorbed through intestine and the drug concentration(C) will have a more gradual rising. These models shown are the areas we explore the possibility of our proposed method.

$$C_{IV} = D(e^{-kt}) \tag{1}$$

$$C_{PO} = D\frac{k_a}{k_a - k}(e^{-k_a t} - e^{-kt}) \tag{2}$$

Drug monitoring is one critical part in personal medicine. However, very few PK model-based approaches have been implemented in this process. The PK modeling data usually stays reserved in clinical and pre-clinical studies while not being utilized in clinical practice. The mere criteria monitored in clinical practice is the drug concentration in patient, where no further PK modeling is performed to reveal the kinetics coefficient [1]. The drug concentration is assumed to be in between the minimum effect and minimum toxicity range. The limitation of this approach is that without having the model implemented, the clinical practitioner cannot foresee the change of patient's physical condition and make a prediction.The clinic on site PK analysis is never a option for the clinical practitioner due to the time and skill-training cost. Therefore, we introduce the concept of supervised machine learning in making PK analysis automated. As previously mentioned, many factors can cause the variation of PK coefficient [2] [3]. By knowing the PK parameters of the drug on each individual, theoretically, the oncoming drug concentration can be predicted and the optimum dosage can be adjusted based on this information.



(a) IV bolous 1 Compartment model (b) IV Onedose (c) IV Multidose

(d) PO oral 1 Compartment model (e) PO Onedose (f) PO Multidose

**x-axis represents time (in hours) and y-axis represents concentration*

Figure 1: Compartment PK Models

2 Related Works

Current PK modeling tools is manual and primitive, and as a result areas of machine learning has not yet been implemented. To determine the type of model and its concentration levels, currently researchers manually retrieve the information from the patient's recorded data of concentration level over time. Each combination of PK model parameters will have its theoretical analytical form, which the recorded data are manually analyzed against. Current methods of calculating these parameters based on the trendline are described in this section.

2.1 Area-Under-Curve

The methods of data analysis for PK models determines a patient's concentration level by measuring the patient's total body exposure to the drug [1]. The method of measurement used takes the area under a curve (AUC), evaluated as

$$F = \frac{D_{iv}(AUC_{0-\infty})}{D(AUC_{iv,0-\infty})} \tag{3}$$

When F less than 1 for a drug administered extravascularly, either the dosage form did not release all the drug contained in it, or some of the drug was eliminated or destroyed (by stomach acid or other means) before it reached the systemic circulation.

The issue here is that the specific method of evaluating AUC has not been specified, and therefore conclusions on the methodology of calculations for this approach is ambiguous. Moreover, the main weakness of this method is in its requirement for manual data analysis using forms of math equations to determine the parameter values. In the real world, the dataset would not be ideal due to inconsistency of multiple variables, such as environment, patient's health condition, etc. Therefore, any variations in the dataset would result in an inaccurate calculation.

2.2 Non-Compartmental Analysis

Improving on the general AUC model above, Gabrielsson and Weiner proposed an approach to approximate the PK model using non-compartmental analysis (NCA). By using a linear trapezoidal approximation per step-interval, the AUC of the model can be calculated using the equation

$$AUC_{t_{last}}^{\infty} = \int_{t_{last}}^{\infty} C_{last} e^{-\lambda_z (t - t_{last})} dt = \frac{C_l ast}{\lambda_z} \tag{4}$$

where $C_l ast$ and λ_z are the last measurable nonzero plasma concentration and the terminal slope on a log_e scale, respectively [4].

NCA reflects the half-life pattern of concentration effect on a patient's body, and therefore is a better representation for evaluating the dosage and concentration levels. However, similar to its primitive approach this method suffers from manual analysis. Large amounts of data would result in significant usage of time during this process. Moreover, a major weakness of NCA is the approximations made in determining the AUC. Though the half-life trend is representative, there are still instances where the estimation for each integral component will overfit or underfit, resulting in measurement error.

2.3 SAS NLMIXED Multi-dose

SAS NLMIXED is a one-compartment model program designed to manipulate data and construct means functions for fitting a patient's data of multiple dosage. Multiple dosage describes the state of a patient's concentration level over time of partaking drug dosage at regular intervals, resulting in a consistent increase of the patients concentration level over a certain period. SAS NLMIXED uses the superposition principle to evaluate and fit the curve [5]. The simplified equation for this measurement is defined as

$$C(t) = \frac{D k_a}{(k_a - k_e) V_D} [\frac{e^{-k_e t}(e^{n \tau k_e} - 1)}{e^{n \tau k_e} - 1} - \frac{e^{-k_a t}(e^{n \tau k_a} - 1)}{e^{n \tau k_a} - 1}] \tag{5}$$

However, there are multiple areas which introduces error in this method. The major issue is that SAS NLMIXED cannot determine the concentration parameters based on the data. Instead, it requires a domain expert to identify the values of the parameters before the data could be fit. The weakness introduces complications in time and resource in finding a domain expert. Furthermore, despite the experience and expertise, there still contains possibility for human error, thus introducing error on the parameters and thereby the results of the fitting. Also, model assumes ideal, regular dosage on constant intervals. In the case the a dose is missed or concentration of the dose varies, the fitting will also fail.

3 Proposed Methods

3.1 Gaussian Proccess (GP) Regression

For the actual PK data, the unknown parameters and models forces use of linear regression with changed basis. Therefore, we chose GP regression as the regression method to interpolate the original sparse data. There are four major non-stationary kernels which can be used here (Radial-basis function (RBF) kernel, Matérn kernel, Rational quadratic kernel, and Exp-Sine-Squared kernel), but the major factor here is to select an efficient kernel. Considering the simplicity in this regression problem (only has a maximum of 4 features), there are four models with different analytical functions. We plotted one figure for each PK model per kernel and manually chose the best kernel for each model. Then, using the chosen kernel for each model to interpolate the original sparse data, we can get more dense data to perform KNN.

3.2 K-Nearest Neighbours (KNN)

The proposed method of prediction used is a non-parametric supervised learning method, KNN. The model will be constructed beforehand with 150 sets of generated PK model data at varying parameters of k, k_a and τ. This process is repeated per each PK model type, resulting in a total of sets of training models. Length of each trained dataset is measured for 48 hours. Predicted parameter labels will be compared against its real label parameters for prediction accuracy. Hyper-parameter of KNN "k" is determined to be 5and the distance will be measured using Euclidean distance. The resulting label for the prediction model can be written as

$$\hat{y} = \Sigma_{i=1}^{5}(z) \tag{6}$$

where z is the minimum 5 values of all calculated distances $||x_i - \hat{x_i}||$.

We decided to use KNN to improve on current PK model data analysis because of KNN's ability to calculate for minor variations. As previously stated, realistic data records of patient's concentration levels may deviate from ideal trends due to various factors. KNN allows the parameters to be determined by the mode of its nearest values. This can potentially predict the parameters more accurately as it allows for some degree of variance reflected in the realistic data.

3.3 DBSDCAN Based Multidose Model Fitting

The problem of the multiple dosage situation is seen during the change of the physical condition [2], like the kidney or liver failure can lead to the change of drug absorption and drug metabolism. Ideally, we expected the drug data to be linear, in which case the absorption coefficient and the elimination coefficient remain the same over the whole drug taking period. However, except for the concerns mentioned above, many factors can have an impact on it. In a real life scenario, although the dosing interval can be well-documented, the lag time which is the actual onset absorption time of the drug absorption may vary. If not taken care it may lead to an inaccurate model fitting and wrong parameter estimation. However, based on the concentration-time graph the true onset of a dose can be determined. Therefore, we developed a clustering based method that can automatically detect the 'onset point'.

By taking in the sparse raw test data, the RBF interpolation method was used to smooth out the curve. This method was performed using the *sklearn* package. Seed points were generated at every 0.1 hour interval. Starting from the seed points, we apply the minimization method with the objective function of the RBF to predict the value. The seed points then converge to the nearest minimum and form a cluster around the local minimum. By applying the DBSCAN method at this stage, we can obtain different cluster that stands for different onset point of a dosage, thereby detecting the dosing interval. The data point is divided into separate dosing interval and perform the PK modeling on each interval. Treat them as a one dose situation. In each dosing interval, we perform the model fitting algorithm using the *scipy* optimization library which as a better smooth function denoted by the equation

$$f(ka, k, T) = Dose * ka/(ka - k) * (e^{-ka*T} - e^{-ka*T}) \tag{7}$$

$$Y = f(ka, k, T) \tag{8}$$

$$Obj(ka, k) = \|\hat{Y} - Y\| \tag{9}$$

4 Experiment

4.1 Gaussian Process (GP) Regression

Based on the method described above, RBF kernel was chosen as the GP regression kernel.

The data in the real clinic is sparse. Typically they will measure the concentration in blood every hour or longer, as discussed in section 1. However those sparse dataset is not sufficient for KNN because the sparse data will have "fake" nearest neighbours when predicting. Methods of fitting used include nonlinear regression analysis or non-compartmental analysis (NCA). For the nonlinear regression analysis, the parameters need to be known. This method is used in kinetics characterization of the pharmacy compound after the parameters are determined. When compared with NCA, GP regression shows significant advantages, resulting in a closer approximation to the real curve as shown in Figure 2.

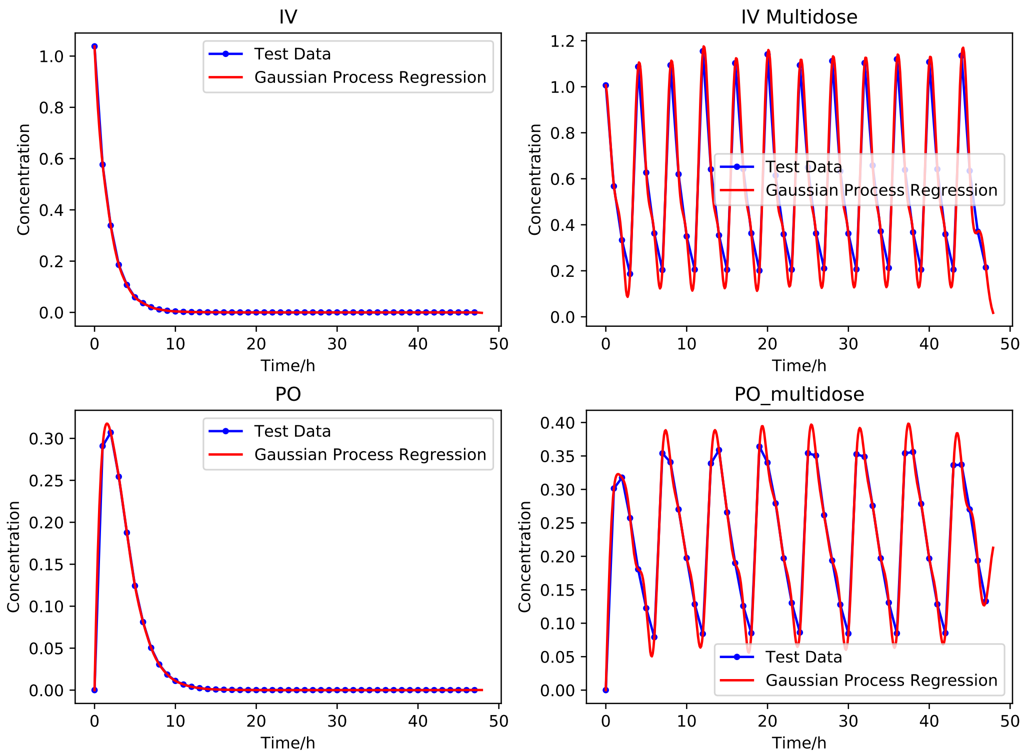


Figure 2: Gaussian process regression (red line) with RBF kernel for 4 different models (IV, IV multidose, PO, PO multidose), and non-compartment modeling (Blue line).

4.2 K-Nearest Neighbours (KNN)

Due to time constraints, evaluation was performed on a total of twelve test datasets (3 for each type of PK model) of varying parameters. The test data was retrieved from published data in [6] [7]. Results of the model prediction accuracy is shown in Table 1.

Drug Name	Amoxicillin Po one dose	Amoxicillin Po multi dose	Oxycodo Po one dose	Oxycodo Po multi dose
Model Type Prediction	Correct	Correct	Correct	Correct
Drug Name	Primidone Po one dose	Primidone Po multi dose	Penicillin IV one dose	Penicillin IV multi dose
Model Type Prediction	Correct	Correct	Correct	Correct
Drug Name	Meperidine IV one dose	Meperidine IV multi dose	Valproicacid IV one dose	Valproicacid IV multi dose
Model Type Prediction	Correct	Incorrect	Correct	Incorrect

Table 1: The Model Type Prediction of the Oral Administrated Drug

Based on the results, we observe that the prediction accuracy of using KNN for for PK model type prediction is 83.33% (further details can be seen in Appendix). Overall, the prediction accuracy is not significant enough for practical application in PK areas. However, further analysis of Table 1 shows that the prediction accuracy for PO-type models predicted 100% while for IV-type models it was 66.67%. This implies that our proposed method is effective in performing predictions for PO-type PK models, while having a less reliable result for IV-type PK models.

Results for parameter prediction accuracy of the test data can be observed in Appendix. Unfortunately, the prediction accuracy for determining the parameters of PK models was negligible, implying that our proposed method was inaccurate predicting specific parameter values.

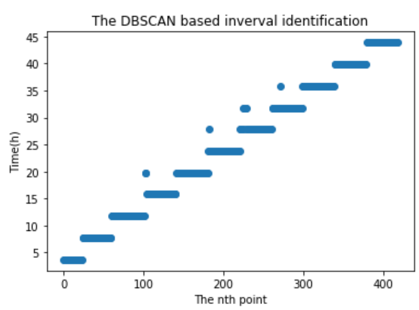
4.3 DBSCAN-Based Multidose Modeling

The test data we used for multidose modeling method was generated using the PK parameter of Oxycodo, an analgesic drug administrated orally with the PK parameter τ(Interval)=8h, $k_a = 1.23, k = 0.58$. The $0 - 48h, 0 - 150h$ data points were generated for The proportional variance [8] was introduced into the model for the test data was introduced:

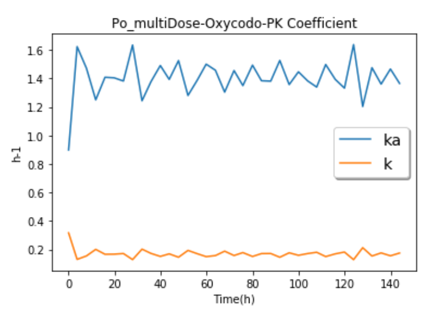
$$Proportional\ Variance : Y = (1 + \sigma) * Y \tag{10}$$

$$\sigma \sim \mathcal{N}(0,\, 0.02)\,. \tag{11}$$

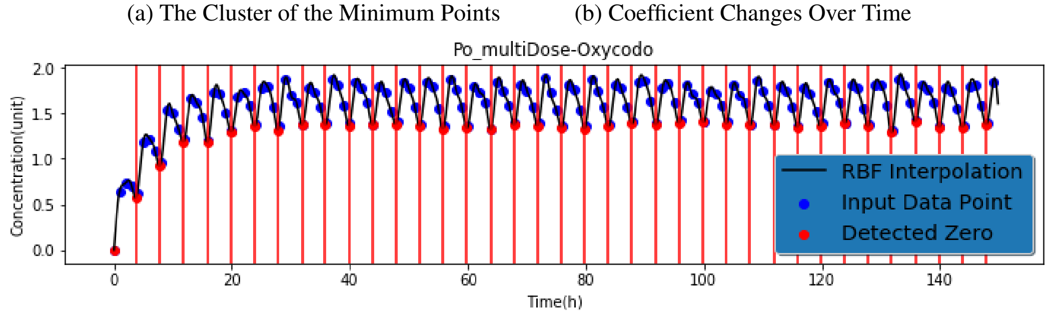
Figure 3(a) is the result of DBSCAN-based multidose interval identification on the simulated data of Oycodo. Parameters $\epsilon = 0.01, n_{minima\_sample} = 20$ were set for clustering, which in this case identified all the intervals and excluded the false positive cluster. Based on the detected intervals, the raw input data points were divided into separate one-dose regimen. The model fitting was performed at each interval to determine the PK coefficient. In our test data, the PK coefficient remains the same over the whole time course. Thus, we would expected the parameters k_a and k to remain the same at every interval. However, due to the loss of accuracy in fitting the model by using small amounts of datasets, there is an expected fluctuation of the estimated coefficient as observed in Figure 3(b).



(a) The Cluster of the Minimum Points



(b) Coefficient Changes Over Time



(c) $0 - 150h$ Dosing Regimen Detected Intervals

(a) Shows the interval identification results from using DBSCAN (b) Shows the coefficients in each interval and the fluctuation over time (c) Shows the input sparse data point and the RBF interpolation of the input data.

Figure 3: DBSCAN-Based Multidose Monitoring

5 Discussion

Our proposed method concludes with accurate predictions in the PK model type while underperforming in model parameter prediction. As shown in the experimental section, the results of KNN is more accurate for PO multidose model compared to IV multidose model. The major difference between the two model types is in the starting stage, while at later periods the models become identical (see Figure 2). Another problem is that, even though KNN can identify the correct model, the parameter cannot be accurately determined. A possible solution would be to develop a simple method to automatically model and find each boundary point. This may produce more accurate results applicable in real time clinical analysis.

Future works for this method is to improve on the prediction accuracy for both PK model type and parameters. One direction could be to add a pre-classifier which can distinguish between the differences of IV and PO models, then use KNN to predict the specific model (onedose or multidose). Another major factor for future works is improvement in the prediction accuracy of the model parameters. This area would require more detailed research for a definite solution.

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6 Appendix

Table 2: The Model Parameter Prediction of the Oral And IV Administrated Drug

Drug Name	Amoxicillin Po multidose			Oxycodo Po multidose			Primidone Po multidose		
Parameter	$k_a(h^{-1})$	$k(h^{-1})$	$\tau(h)$	$k_a(h^{-1})$	$k(h^{-1})$	$\tau(h)$	$k_a(h^{-1})$	$k(h^{-1})$	$\tau(h)$
Parameter Value	0.77	0.58	8	0.3	0.15	4	0.005	0.134	24
Prediction Value	0.112	0.6	8	0.352	0.04	11.5	0.122	2	9

Drug Name	Amoxicillin Po onedose		Oxycodo Po onedose		Primidone Po onedose	
Parameter	$ka(h^{-1})$	$k(h^{-1})$	$k_a(h^{-1})$	$k(h^-1)$	$k_a(h^{-1})$	$k(h^{-1})$
Parameter Value	0.77	0.58	0.3	0.15	0.005	0.134
Prediction Value	652	0.02	0.152	4.9	0.012	0.4

Drug Name	Penicillin G IV multidose		Meperidine IV multidose		valproic acid IV multidose	
Parameter	$k(h^{-1})$	$\tau(h)$	$k(h^{-1})$	$\tau(h)$	$k(h^{-1})$	$\tau(h)$
Parameter Value	2.34	4	0.0647	6	0.053	12
Prediction Value	1.3	4	0.06	6	0.1	3.5

Drug Name	PenicillinG IV onedose	Meperidine IV onedose	valproic acid IV onedose
Parameter	$k(h^{-1})$	$k(h^{-1})$	$k(h^{-1})$
Parameter Value	2.31	0.0647	0.053
Prediction Value	1.9	Model Error	Model Error